

Farah Reda Farahat Sanad

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Professional Summary

Detail-oriented and technically proficient instructor and developer with solid experience in database systems, machine learning, front-end development, and computational task optimization. Adept at designing, building, and deploying full-cycle projects with a user-focused approach. Proven ability to deliver educational content and coach students in both academic and competitive environments.

EDUCATION

Computer Science and Information Technology, Artificial Intelligence and Data Science section, Egypt
Japan University of Science & Technology (E-JUST) [2022 - 2026]

WORK EXPERIENCE

Private Math Instructor - Freelance

2024 — Present

- Delivered tailored tutoring in Linear Algebra and Discrete Mathematics to undergraduate students.
- Simplified complex concepts through problem-based teaching strategies.

Certified Coach & Instructor First Lego League Challenge 2021 — Present

- Coached and mentored student teams in robotics, teamwork, and innovation projects.
- Delivered engaging sessions to enhance students' problem-solving and engineering skills.

Internships

2025 — Present

- **Data Analysis Trainee at NTI - National Telecommunication Institute**
 - Gained hands-on experience in data cleaning, visualization, and analysis using Python (pandas, matplotlib) and SQL.
 - Worked on real datasets to extract insights, build dashboards, and present findings.
- **Web Development Trainee at ITI -Information Technology Institute**
 - Trained in front-end and back-end web technologies including HTML, CSS, JavaScript, React.js, and Python.
 - Developed small-scale web applications and worked on responsive design and API integration.

PROJECTS

Parallel Task Execution Web App

- Developed a responsive web application that allows users to select and execute specific computational tasks.
- Technologies used: MPI (Open MPI), Python, HTML, CSS, JavaScript.
- Designed a dynamic UI and integrated real-time task execution through backend parallelization using MPI.
- Optimized communication between front-end and backend processes for performance efficiency.

SKILLS

Programming Languages:

- Proficient in C++, Python (OOP and problem-solving)

Web Development:

Front-end: HTML, CSS, JavaScript (Basic)

- Backend Integration: Python with Web App Frameworks

Database Systems:

- Design & Development using SQL and Relational Models
- Strong grasp of ERD modeling, Normalization, and Query Optimization
- Data Science & AI:
 - Data preprocessing and Feature Engineering
 - Machine Learning: Classification, Regression, Clustering
 - Visualization tools: Power BI for interactive dashboards and reporting
 - Integration of AI models into responsive web apps

LANGUAGES

- Arabic: Native
- English: C1
- Japanese: A2